

(Please write your Enrollment Number)

Enrollment No. _____

REMID-TERM EXAMINATION
(Course Name: B. Tech. (ECE, ECE AI-1 & ECE AI-2) (Semester: 1st)
(November 2024) OFF LINE mode

Subject Code: BEC 103	Subject: ELECTRONICS WORKSHOP
Time : 1 ½ Hours	Maximum Marks :30

Note: Q1 is compulsory. Attempt any two parts of Q2 and Q3

Q1		(2.5*4=10)	CO Mapping
a)	Explain different types of semiconductor diodes.		UNIT 1 & 2
b)	Explain the working of Zener diode. Can you use it as voltage regulator? Explain.		
c)	Explain the working of n-p-n type bipolar junction transistor. Explain w.r.t two diodes connected back to back? Explain.		
d)	What is clipper circuit? Explain the circuit functionality.		
UNIT I			CO Mapping
Q2	Attempt any two parts	(5*2=10)	
a)	Explain capacitances in depletion and diffusion mode of a PN junction semiconductor. Write down its formulae and explain.		UNIT 1
b)	Explain forward bias and reverse biased in PN junction diode. Explain break down region. Can you utilize breakdown region in any way? Explain		
c)	Explain CB configuration. Explain operating point, DC load line and Q Point.		
UNIT II			CO Mapping
Q3	Attempt any two parts	(5*2=10)	
a)	How can dope a transistor. Explain with NPN and PNP transistor both.		Unit 2
b)	Explain all types of stabilization in BJT/NPN transistor? What is the need of it?		
c)	An a.c. voltage of peak value 20V is connected in series with a silicon diode and load resistance of 500 Ω . If the forward resistance of diode is 10 Ω , find: (i) peak current through diode (ii) peak output voltage. What will be these values if the diode is assumed to be ideal??		